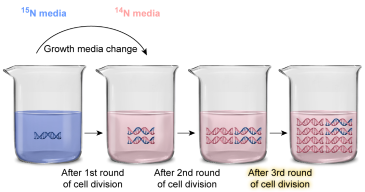
Escherichia coli bacteria containing only ^{15}N-labeled DNA were grown in media containing only ^{14}N nucleotides. What percentage of double helices would be composed of one ^{15}N strand and one ^{14}N strand after three rounds of cell division?

- A. 12.5% [31%]
- ✔ B. 25% [45%]
- C. 50% [14%]
- D. 100% [8%]

Correct
44% answered correctly

Explanation:

Semiconservative DNA replication



Generation	P	F1	F2	F3
DNA with two ^{15}N strands	100%	0%	0%	0%
DNA with one ^{15}N strand & one ^{14}N strand	0%	100%	50%	25%
DNA with two ^{14}N strands	0%	0%	50%	75%

DNA replication is a semiconservative process in which DNA polymerase synthesizes each daughter strand using one of the parental strands as a template. Each new double helix contains one template **parental strand** and one newly synthesized **daughter strand**.

In the experiment described, all DNA strands in the *Escherichia coli* bacteria initially have the heavy isotope ^{15}N in their DNA. After the first round of cell division, no double helices with two parental ^{15}N strands remain because these will all have separated during cell division. Instead, the double helical DNA of *all* (100%) the offspring (F1 generation) will consist of **one inherited parental** strand with ^{15}N and **one daughter strand** with ^{14}N **(Choice D)**.

During the second round of cell division, **the F1 double helices separate to become templates** for production of the F2 generation. Because all F1 DNA double helices contain one ^{15}N strand and one ^{14}N strand, *exactly half* of the template *strands* contain ^{15}N DNA. Therefore, *half* of the offspring (50%) will contain one inherited ^{15}N strand and one newly synthesized ^{14}N strand **(Choice C)**. The other half of the offspring (ie, those that inherited a ^{14}N parental strand) strand will have *two* ^{14}N strands in their double helix —one inherited and one newly synthesized.

Similarly, for each new generation the percentage of *double helices* containing one ^{15}N strand and one ^{14}N

strand will be **half the percentage of the previous generation**. Therefore, **after the third round of cell division, 25%** of double helices contain one ^{15}N strand and one ^{14}N strand.

(Choice A) Although 12.5% of individual DNA *strands* contain ^{15}N after three rounds of division, the question asks about *double helices* containing *both* ^{15}N *and* ^{14}N strands. A value of 12.5% would be the percentage of double helices that contain both a ^{15}N strand and a ^{14}N strand after *four* rounds of cell division (F4).

Educational objective:

DNA replication is a semiconservative process that results in each double helix containing one parental strand and one newly synthesized daughter strand. DNA polymerase synthesizes each new daughter strand by using a parental strand as a template.

Time spent: 0 sec	**QID**:402144
Subject: Biology	**System:** Molecular Biology